RBDN: A Robust Background Denoising Network for Weakly Supervised Temporal Language Grounding

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***Abstract*—Temporal Language Grounding (TLG) with weak supervision aims to retrieve events in untrimmed videos corresponding to text queries using only video-text pairs as annotations. However, noisy background semantics in videos lead to inconsistencies in cross-modal alignment, which further hinder the grounding of query events. To address this, we introduce a Robust Background Denoising Network (RBDN), which refines backgrounds and mitigates their negative impact on events. RBDN leverages robust PCA and Frequency Augmentation techniques to filter out spatially static and temporally regular background information, thereby denoising the undesired gradient influence. Extensive experiments on well-known datasets, such as Charades-STA and ActivityNet-Captions, demonstrate that our approach significantly outperforms current benchmarks.**

***Index Terms*—weakly supervised temporal language grounding, robust PCA, frequency augmentation, multi-modal representation learning**

I. Introduction

In computer vision, widespread attention has been given to the understanding of video content, e.g., Multiple Object Detection [1], Person Re-identification [2], Video Anomaly Detection [3], and etc. As an increasingly attractive task, Temporal Language Grounding (TLG) [4] aims to ground the moments of target events that people focus on within untrimmed videos, specifically, the starting and ending frames. Orthogonal to existing tasks, e.g., temporal action localization [5] or action recognition tasks [6] in computer vision, TLG, being a video retrieval task, not only requires processing the video itself but also incorporates information from other modalities to enhance the understanding of video content, thereby consequently grounding the start and end frames of events. Traditional TLG methods are under full supervision, requiring datasets to provide precise timestamp-level annotation. However, these human annotations are resource-intensive and often challenging to obtain. Furthermore, timestamp-level annotations are subject to significant variations based on the annotators, leading to the undesired inconsistency of the annotations. Accordingly, weakly supervised TLG tasks have become the predominant research focus in recent years.

However, conventional benchmark weakly supervised methods cannot accurately localize the events because a significant deviation exists between the video segments in the visual modality and the corresponding queries in the textual modality. Thus, learning representations of the different modalities to guarantee consistent self-supervision comes to be the most challenging problem of weakly supervised TLG methods. To intuitively understand the intrinsic reason behind such a problem, we conduct an empirical exploration of real-world TLG datasets. As demonstrated in Figure 1, we observe from real-world video samples and explore that *events* with specific semantics are usually present in a prominent position, which is spatially and temporally consistent, while the other parts of videos can be regarded as the *background*. Thus, the background semantics can be treated as a confounder during the training of weakly supervised TLG models, degenerating representations' training efficiency and discriminability.



Fig. 1. The empirical observation determining the discrimination between *events* and *backgrounds* concerning the spatial and temporal consistency on the video data.

In this paper, to reduce the negative influence of the noisy background semantics during the cross-modal alignment, we follow our empirical exploration and further propose a ***R**obust **B**ackground **D**enoising **N**etwork* (*RBDN*), leveraging the insight of background denoising towards input video samples for weakly supervised TLG. Compared with the event semantics, the background semantics exhibit the characteristics of spatially static and temporally regular. Thus, we innovatively apply robust PCA techniques to decompose the dynamic and static information in the visual input, filtering out the

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static information. We further project video samples into the frequency domain to filter out the vector dimensions that are significantly distributed, thereby avoiding the undesired gradient influence on model training of the background.

The major **contributions** of this paper are three-fold:

- We have discovered that in weakly-supervised TLG tasks, background information significantly contributes as noise to both video and text modalities. We also redefine the spatiotemporal characteristics of background information: spatially static and temporally regular in variation.
- We proposed a novel Robust Background Denoising Network (RBDN) to denoise the background semantics from spatial and temporal perspectives by using robust PCA and frequency augmentation.
- Experiments show that RBDN outperforms recent well-established methods, proving that our analysis of the existing weakly supervised TLG methods and the application of visual background denoising are effective and forward-looking.

II. Related works

Weakly supervised TLG Most existing weakly supervised TLG methods [7]–[11] adopt a Multi-Instance Learning (MIL) framework, which generally regards the input video as a bag of instances with bag-level annotations. Therefore, MIL based methods typically use videos that do not match the batch as negative instances to strengthen cross-modal correlation. CCL [12] and VCA [13] firstly introduce contrastive learning mechanisms to effectively distinguish positive and negative proposals. Another prevalent strategy in weakly supervised TLG involves reconstruction-based approaches. For instance, SCN [14] and ACN [15] generate several candidate segments within a video, selecting the top segments capable of reconstructing the masked terms within a given natural language query. This method leverages the reconstruction to identify informative video segments related to the textual description.

To go further with contrastive learning, some weakly supervised TLG methodologies introduce innovative sampling techniques. CNM [16], for instance, incorporates Gaussian sampling for effective mining of positive and negative samples. This technique enhances feature discernment by strategically selecting samples that contribute meaningfully to learning.

Building upon these concepts, CPL [17] employs distinct Gaussian sampling to generate positive and negative proposals. This advanced strategy further augments the efficiency of sample mining, allowing the model to focus on key areas of the video that contribute significantly to understanding the event described in the query.

Robust PCA Principal Component Analysis (PCA) is one of the most widely used tools for data analysis and dimensionality reduction. Over the years, several approaches have been developed to make PCA more robust, including influence function techniques [18], [19], multivariate trimming [20], alternating minimization [21], and random sampling methods [22]. More recently, [23] introduced Principal Component Pursuit, a convex optimization method that enables the exact recovery of both low-rank and sparse components. Building on this, [24] extended the approach to recover both low-rank and sparse components in 3-dimensional tensors.

Frequency augmentation. Timesnet [25] proposes a method of projecting temporal data into the frequency domain via a fast Fourier transform, selectively extracting periodic components within the temporal data. This work inspires us to adopt frequency-domain augmentation methods to denoise background noise, such as camera shaking or light flickering, which are difficult to eliminate in the original video.

III. Methodology

Figure 2 illustrates the pipeline for our proposed RBDN. In the following section, we will present a comprehensive breakdown of the intricate structure of our proposed method.

A. Feature Extraction

Given an untrimmed video v and its query sentence t, we extract their visual feature $v^e = E_v(v) \in V^{N\times L_v\times D_v}$ and textual feature $t^e = E_t(t) \in T^{N\times L_t\times D_t}$ by fixed pre-train models $E(\cdot)$. Here, the symbol N represents the total number of input samples. Furthermore, L_v and D_v correspond to the temporal length and dimension of the encoded visual features, and L_t and D_t denote the word length and dimension of the encoded textual features, respectively.

B. Robust Background Denoising

In this work, we focus on denoising the background semantics in visual features that hinder the effective fusion of textual queries with event semantics. In TLG tasks, background information in videos exhibits two distinct characteristics:

- Spatially static which can be attributed to the low-rank structure of background semantics across the entire video, stemming from the inherent correlation between frames.
- Temporally regular which often result from challenging recording conditions, e.g., camera jitter or flickering.

Tensor Robust PCA. In order to effectively denoise spatially static tensors representing background semantics, we employ RPCA for visual tensors. Specifically, given a visual tensor v^e, it can be decomposed as:

$$v^e = \mathcal{L}_0 + \mathcal{E}_0, \tag{1}$$

where $\mathcal{L}_0$ is low-rank and $\mathcal{E}_0$ is sparse, and both components are of arbitrary magnitudes. The decomposition is typically achieved by minimizing the following objective function:

$$\min_{\mathcal{L},\mathcal{S}} \|\mathcal{L}\|_* + \lambda\|\mathcal{S}\|_1 \quad \text{subject to} \quad v^e = \mathcal{L} + \mathcal{S} \tag{2}$$

where $\|\mathcal{L}\|_*$ is the nuclear norm of $\mathcal{L}$, promoting low-rank solutions, and $\|\mathcal{S}\|_1$ is the l_1-norm of S, encouraging sparsity. The regularization parameter λ controls the trade-off between rank minimization and sparsity.

To solve this problem, the alternate minimization strategy is often employed. The algorithm alternates between updating $\mathcal{L}$ and $\mathcal{S}$ using appropriate optimization techniques, such as singular value thresholding for $\mathcal{L}$ and soft-thresholding for

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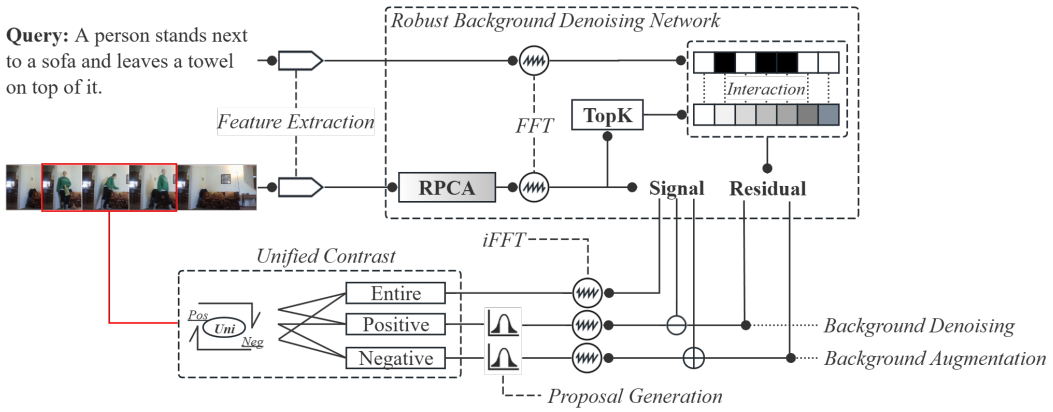


Fig. 2. First, features of the query and video are extracted, followed by RPCA decomposition of the video features. RPCA denoises spatially static components from the background semantics. Subsequently, features from both modalities are projected into the frequency domain to further eliminate temporally regular background semantics. Regarding the background as a residual, the event information can be detected and further augmented. Subsequently, we obtained three types of grounding proposals through the proposal generator. Employing the unified contrast, we ultimately achieved precise localization of event semantics within the video segments.

$\mathcal{S}$. These updates aim to iteratively minimize the objective function, ensuring that the low-rank and sparse components are appropriately balanced. The core challenge is handling the multi-dimensional structure of the tensor, which often requires specialized algorithms like higher-order singular value decomposition (HOSVD) [26] for efficient optimization.

Frequency Augmentation. To denoise background semantics with temporally regular characteristics, we no longer focus solely on inducing the fusion of visual and textual features in their original sequences and employ the FFT operator to transfer $\mathcal{S}$ and t^e from the two distinct modalities into the frequency domain. For visual modality, As shown in Figure 3, we employ top-k sampling to extract frequently occurring semantics, identified as dimensions with higher amplitudes in the frequency domain. These background semantics are treated as residuals:

$$v^f = \mathit{FFT}(\mathcal{S}) - \mathit{TopK}(\mathit{FFT}(\mathcal{S}), k) \ . \tag{3}$$

$$t^f = \mathit{FFT}(t^e) \ . \tag{4}$$

To ensure a specific and robust alignment between v^f and t^f, employ cosine similarity for cross-modal interaction and utilize a 3-layer fully connected (FC) network to bring learnable parameters.

Subsequently, we employ an inverse FFT (iFFT) operation to convert the frequency feature back to the temporal domain.

C. Gaussian Grounding

This section will introduce our unified multi-scale contrastive grounding network that utilize Gaussian models to generate grounding proposals.

Proposal Generation. Up to this point, our next objective is to identify video segments that most closely align with the query semantics by grounding the combined features within the video sample. Following CPL [17], as $\hat{v}$ represents the fused video features after iFFT, we utilize a θ-dimensional Gaussian distribution to predict positive masks M^p, and generate positive proposals p^p:

$$p^p = M^p \cdot \hat{v} \ . \tag{5}$$

Unlike CPL, which uses the remaining parts of positive proposals as negative proposals, we incorporate negative sampling to generate negative proposals. Specifically, we perform reverse augmentation on video samples by adding background semantics to the video features, producing $\hat{v}'$, and then generate negative proposals p^n:

$$p^n = M^n \cdot \hat{v}' \ . \tag{6}$$

To ensure that the θ positive proposals are distinct, we aim to maximize the entropy of the generated mask centers, thus allowing the θ proposals to carry a more significant amount of information.

$$l_{ent} = -\sum_{i=1}^{\theta} M_i log(M_i) \ . \tag{7}$$

In addition, we take an entire mask as a reference proposal p^r for each video sample. In this context, for each video sample, among the θ positive proposals that have already been generated, our objective is to select the proposal of the top-1 cosine similarity p^{p1} to the text query. To achieve this, we employ an inner-sample grounding loss l_g:

$$l_g = \frac{1}{N}\sum_{i=0}^{N}(S_c(p_i^{p1}, t_i^e) + S_c(p_i^r, t_i^e)) \ , \tag{8}$$

where we calculate the cross-modal similarity $S_c(\cdot)$ by a cross entropy loss. Here, considering that in real-world datasets,

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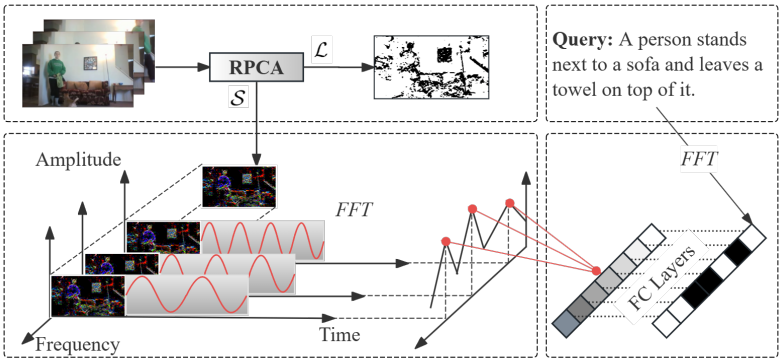


Fig. 3. When performing FFT on video features, we obtain a spectrum where the independent variable is the sampling frequency, and the dependent variable consists of the real part as amplitude and the imaginary part as phase. Analyzing the amplitude, we can identify the top K frequencies and treat them as background semantics. After interacting with the textual features projected into the frequency domain, these background semantics are treated as residuals and denoised from the video features.

there are conditions that the query of a video sample corresponds to the entire video, the entire video must be a proposal. So we also add p^r in the l_g to cover such situations.

Unified Contrast. In contrastive learning, it is a general goal to maximize the within-class similarity S_p and minimize the between-class similarity S_n. Here, we employed a unified loss l_{uni}:

$$l_{uni} = log(1 + \sum_{i=1}^{\theta} \sum_{j=1}^{N} exp(\gamma(a_n^j S_n^j - a_p^i S_p^i)) \quad . \tag{9}$$

Here, a_p^i and a_n^i are self-attention parameters to adjust the weight of similarity values, while S_p and S_n are calculated by cross-entropy losses.

Finally, taking (p^p, p^r), (p^p, p^n) and (p^r, p^n) as pairs of contrastive learning, we have the multi-scale loss l_{ms}:

$$l_{ms} = l_{uni}(p^p, p^r) + l_{uni}(p^p, p^n) + l_{uni}(p^r, p^n) \quad . \tag{10}$$

D. Loss function

Our proposed RBDN includes three parts of loss: An entropy loss l_{ent} that ensures the distinctness of proposals from others, a grounding loss l_g to enhance the accuracy of grounding, and a unified multi-scale contrastive loss l_{ms}:

$$l_{total} = l_{ent} + \alpha_1 l_g + \alpha_2 l_{ms} \tag{11}$$

where α_1 and α_2 are hyperparameters of balancing.

IV. Experiments

To fairly evaluate the performance of RBDN approach in TLG tasks, drawing on the existing sota approach with the existing TLG domain, we conducted performance experiments on RBDN using public datasets Charades-STA and ActivityNet-Captions and used IoU as an evaluating metric for measuring the TLG effectiveness of RBDN.

A. Experiment settings

A typical TLG dataset comprises a set of videos, each accompanied by one or more annotations, forming moment-query pairs. In this context, each annotation includes a query corresponding to a specific moment within the video. For weakly supervised TLG tasks, we solely utilize pairs consisting of video samples and their corresponding queries without relying on annotations for specific moments.

To assess the performance of our proposed methodology, we conducted a series of experiments on two widely recognized public datasets: Charades-STA [4] and ActivityNet-Captions [27]. A detailed introduction of these two datasets is introduced in Appendix I.

In alignment with precedents set by previous research, we adopt the "$R@n, IoU@m$" evaluation metric. The "$R@n, IoU@m$" is the percentage of queries having at least one result whose temporal IoU with ground truth is larger than m among the top-n retrieved moments. For query q_i, among its top-n retrieved moments, if there exists at least one moment whose IoU with ground truth is larger than m, then q_i is considered as positive, denoted by $r(n; m; q_i) = 1$. Otherwise, $r(n; m; q_i) = 0$. Thus, "$R@n, IoU@m$" is calculated as:

$$R@n, IoU@m = \frac{1}{N_q} \sum_{i=1}^{N_q} r(n, m, q_i) \tag{12}$$

Specifically, we set $n = 1$ and $m = \{0.3, 0.5, 0.7\}$ for the Charades-STA dataset. For the ActivityNet-Captions dataset, we adjusted the parameters to $n = 1$ and $m = \{0.1, 0.3, 0.5\}$.

B. Implementation details

To emphasize the contribution of our proposed RBDN in modality fusion, we utilized the same pre-trained network as the most existing state-of-the-art methods for feature extraction from visual and textual inputs. Specially, we take the pre-extracted I3D [28] visual features for Charades-STA with 512 dimensions and C3D features for ActivityNet-Captions with 500 dimensions as the visual input. Conversely, as for the

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TABLE I
EVALUTION RESULTS ON THE CHARADES-STA DATASET. THE NUMBERS IN BOLD ARE THE BEST RESULTS.

Method	IoU=0.3	IoU=0.5	IoU=0.7
TGA [7]	32.14	19.94	8.84
SCN [14]	42.96	23.58	9.97
WSTAN [10]	43.39	29.35	12.28
VLANet [29]	45.24	31.83	14.17
LoGAN [9]	48.04	31.74	13.71
MARN [30]	48.55	31.94	14.81
WSRA [31]	50.13	31.20	11.01
VCA [13]	58.58	38.13	19.57
LCNet [32]	59.60	39.19	18.87
RTBPN [11]	60.04	32.36	13.24
CNM [16]	60.39	35.43	15.45
CPL [17]	66.40	49.24	22.39
RBDN (ours)	**67.54**	**50.82**	**24.40**

TABLE II
EVALUTION RESULTS ON THE ACTIVITYNET-CAPTION DATASET. THE NUMBERS IN BOLD ARE THE BEST RESULTS.

Method	IoU=0.1	IoU=0.3	IoU=0.5
WS-DEC [33]	62.71	41.98	23.34
VCA [13]	67.96	50.45	31.00
EC-SL [34]	68.48	44.29	24.16
MARN [30]	-	47.01	29.95
SCN [14]	71.48	47.23	29.22
BAR [35]	-	49.03	30.73
RTBPN [11]	73.73	49.77	29.63
CTF [36]	74.2	44.3	23.6
WSLLN [8]	75.4	42.8	22.7
LCNet [32]	78.58	48.49	26.33
CCL [12]	-	50.12	31.07
WSTAN [10]	79.78	52.54	30.01
CNM [16]	78.13	55.68	33.33
CPL [17]	79.86	53.67	31.24
RBDN (ours)	**80.43**	**56.14**	**33.47**

textual input, we apply the pre-trained GloVe model to extract word embeddings with 300 dimensions.

Diving into RBDN, the regularization papameter λ of TRPCA is set 1, and the number of top-k sampling in frequency augmentation is set to 50. As we implement the cross-modal interaction in the frequency domain by a 3-layer full connection network, the output size of them are set 256.

Then, in the part of gaussian grounding, the regualarization parameter γ in Equation 9 is set 256, and the balance parameter α_1 and α_2 are set 1 as a default in Equation 11. Adam optimizer is adopted during training with a learning rate $lr = 0.0004$.

C. Comparisons to the State-Of-the-Art

Table I and Table II illustrate the performance comparison of previous state-of-the-art weakly supervised TLG works evaluating Charades-STA and ActivityNet-Captions.

Table I presents a performance comparison of previous state-of-the-art weakly supervised TLG methods evaluated on the Charades-STA dataset. The results indicate that RBDN achieved the best performance across all evaluation metrics. This underscores the superiority of RBDN in weakly supervised TLG tasks. Notably, in the "IoU@0.7" metric, RBDN outperformed state-of-the-art methods by over 2%, demonstrating its ability to precisely separate background semantics in video samples under weakly supervised conditions and achieving excellent event semantics grounding results.

On the other hand, Table II provides a performance comparison of previously state-of-the-art methods evaluated on the ActivityNet Caption dataset. The results also indicate that our proposed RBDN achieved the best performance across all "R@1" evaluation metrics. This further substantiates that RBDN can maintain excellent performance on a large-scale dataset, affirming its advantages in terms of generalization.

The results on both datasets show that our proposed RBDN exhibits greater performance advantages on finer-grained evaluation metrics. Through retrospective analysis of this result, we identified the cause: RBDN can achieve better grounding boundaries for ambiguous video segments, which is visually demonstrated by the qualitative results in Appendix II.

In addition to the "R@1" metric, we also used the "R@5" metric to validate the performance of RBDN. Compared to "R@1", the "R@5" metric allows for some margin of error, providing a broader pathway for evaluating the model's performance. The complete experimental results are provided in Appendix III.

TABLE III
ABLATION STUDY ON THE COMPONENTS OF RBDN. THE TABLE EVALUATES THE CONTRIBUTION OF EACH MODULE, INCLUDING RPCA AND FREQUENCY AUGMENTATION, TO THE OVERALL PERFORMANCE.

RPCA	V	T	IoU=0.3	IoU=0.5	IoU=0.7
✓	✓	✓	**67.54**	**50.82**	**24.40**
	✓	✓	64.00	48.96	23.21
✓		✓	64.19	48.82	22.04
✓	✓		66.66	49.11	22.70
✓			63.55	47.25	21.91

D. Ablation study

To comprehensively assess the effectiveness of the Robust Background denoising mechanism, we adopt the complete "R@1" metrics to construct our ablation experiments on the Charades-STA dataset. Specifically, we employed a simple 3-layer, fully connected Multi-Layer Perceptron (MLP) as the baseline model.

As shown in Table III, the notations in the first column indicate whether RPCA is used, while those in the second and third columns show if frequency augmentation is applied to the visual or textual modality. Compared to the baseline models, TRPCA proves effective, achieving the best results when augmentation is applied to both modalities.

V. CONCLUSION

This work proposed an innovative Robust Background Denoising Network (RBDN) under a weakly supervised paradigm. By projecting spatiotemporal features of video samples into the frequency domain, RBDN effectively sieves out the regularly patterned background semantics. This filtration enhances the extraction of event features within untrimmed video samples through noisy backgrounds. Furthermore, utilizing a multi-scale unified contrastive loss function significantly

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preserving the consistency of cross-modal alignment within the weakly supervised TLG framework. The robust performance of the proposed method across two widely recognized datasets substantiates the feasibility and potential of RBDN.

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